SELVAGANESH V

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Artificial Intelligence & Machine Learning

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SUMMARY

AI & ML graduate (B.Tech, 2026) who has designed, built, and deployed three production AI applications end-to-end: a FAISS-based RAG document assistant with top-3 passage retrieval and hash-based caching, a multilingual RoBERTa sentiment classifier, and an Azure OpenAI vision app for crop identification. Comfortable shipping full-stack ML products with Python, PyTorch, Hugging Face Transformers, and Gradio/PWA deployment.

SKILLS

Languages: Python, JavaScript, HTML/CSS

AI/ML: Hugging Face Transformers, PyTorch, Sentence-Transformers, FAISS, NumPy, RAG

APIs & Backend: Azure OpenAI, OpenRouter, Groq, FastAPI, Flask, Uvicorn, Gradio, PyMuPDF

Frontend & Cloud: PWA, service workers, Docker, Git, Vercel, Hugging Face Spaces

Languages Spoken: Tamil (Read, Write, Speak), English (Read, Write, Speak)

PROJECTS

PDF-RAG Application

GitHub | Live App

- Built a PDF question-answering assistant in **Python** using **PyMuPDF** text extraction and **Sentence-Transformers** (`all-MiniLM-L6-v2`) embeddings indexed in **FAISS**, to retrieve the passages relevant to each query.
- Grounded answers by passing the **top-3 FAISS** matches as context to an **OpenRouter** LLM, so replies cite the uploaded document rather than model memory.
- Added hash-based caching over embeddings and queries with an automatic **5-minute** cache refresh cycle, so repeated questions return without re-running the retrieval pipeline.
- Deployed a **Gradio** interface on **Hugging Face Spaces** with multi-PDF upload and asynchronous processing, capping context at **15,000 characters per source** so large documents index without blocking the UI.

Emotion Detection in Text

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- Built a sentiment analyser on the pre-trained RoBERTa checkpoint **cardiffnlp/twitter-roberta-base-sentiment**, running **PyTorch** inference with a **512-token** max sequence length to label text Positive, Neutral or Negative with a confidence score.
- Added **langdetect** detection and automatic translation to English ahead of inference, extending a single English-only **3-class** model to any input language.
- Exposed the model as a **Gradio** backend API and consumed it from the browser through the Gradio JS client, with a 500-character input cap and in-page results.
- Built an installable **PWA** frontend in vanilla **JavaScript** with a service worker and deployed it on **Vercel**, so the tool runs from a phone home screen and works offline.

Crop Identification System

GitHub | Live App

- Built a crop-identification web app using the **Azure OpenAI** vision model over **base64**-encoded uploads, to name the crop in a photo and answer follow-up farming questions.
- Preprocessed images with **PIL** (resize, RGB conversion, JPEG compression) to keep vision-API payloads under **500 KB** and response times low.
- Kept session memory across turns, so an identification result carries into a multi-turn farming-advisor chat instead of restarting each question.
- Shipped a no-build vanilla **HTML/JavaScript PWA** on **Vercel** with drag-and-drop upload and offline support, with API credentials loaded from a `.env` config.

CERTIFICATIONS & TRAINING

Python for Scientific Computing – freeCodeCamp — 2025 | Certificate

Generative AI Workshop – NXT Wave — 2024 | Certificate

EDUCATION

B.Tech Artificial Intelligence & Machine Learning (Lateral Entry, 2nd year, post-Diploma) — 2022–2026
Bannari Amman Institute of Technology, Sathyamangalam — CGPA: 7.51

Diploma in Electrical & Electronics Engineering — 2020–2023
PSG Polytechnic College, Coimbatore — 83.41%